Open Source Formal Verification

Introduction/Content

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Who?

- Professor: Yann Thoma
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Short CV

- 2001: Master's degree in computer science from EPFL
- 2005: PhD from EPFL (reconfigurable systems)
- 2005–2009: Lecturer at EIG (now HEPIA; digital systems and processor architecture)
- 2006–2008: Engineer for the Group of Applied Physics (UniGe)
- 2006–2008: Engineer at REDS (HEIG-VD)
- 2009– : Professor at REDS
  - Teaching digital systems, verification of embedded systems, concurrent programming and realtime programming
  - Various research projects (FPGA development, embedded systems, pharmacology)
- 2015–2018: Director of REDS
- 2018–2019: Visiting professor at UNSW
- 2025: Same training given at CERN

My goals here

- Give you a good overview of formal verification
- Show you how to use PSL to write assertions and coverage goals
- Show you Open Source tools work quite well for formal verification
- Let you practice with exercises and examples
  - The best way to learn is to practice!
- Develop a community of CERN engineers who can use formal verification in their work
- Have fun and enjoy the course

Who are you?

Training Material

You can get all the material on the following git repo:
`https://github.com/ythoma/formal-cern-26`

- Theory
  - Slides
  - Code examples
- Exercises
  - Statements
  - Code

Course Schedule

- 3 days of training
- 3.5 hours of theory per day
- 3.5 hours of exercises per day
- 30 minutes of breaks per day
- 1 hour of lunch per day
- The next detailed schedule is likely to shift a bit

Schedule - day 1

Time	Description	Type
9:00	Introduction	Theory
9:15	Introduction to verification	Theory
10:30	Break	
10:45	Introduction to formal verification	Theory (and an exercise)
12:00	Basic PSL constructs	Theory
12:30	Lunch	
13:30	SBY	
13:45	Combinational systems	Exercise
15:15	Break	
15:30	Temporal logic	Theory
16:30	Combinational systems	Exercise
17:30	End of the day	

Schedule - day 2

Time	Description	Type
09:00	Temporal logic	Theory (end)
09:30	Verification of a counter	Exercise
10:00	Verification of a timer	Exercise
10:45	Break	
11:00	Verification of a state machine	Exercise
12:30	Lunch	
13:30	Regular expressions	Theory
15:00	Regular expressions	Exercise
15:30	Break	
15:45	Regular expressions	Exercise
17:15	Parametric design	Theory
17:30	End of the day	

Schedule - day 3

Time	Description	Type
09:00	Coverage	Theory
09:15	Coverage	Exercise
10:15	Helper code and variables	Theory
10:45	Break	
11:15	Backends	Theory
11:45	Backends	Exercise
12:00	Sharing	Theory
12:30	Lunch	
13:30	Backends	Exercise
14:00	Continuous integration and scripts	Theory
14:30	Methodology insights	Theory
14:45	Use cases	Exercise and discussions
15:30	Break	
15:45	Use cases	Exercise and discussions
17:20	Feedback	
17:30	End of the day	